

Lecture 7: Ridge and Lasso Regression, Density Estimation

ML/AI for Engineering Applications
ENV 6932

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Outline

- Recall: Ridge and Lasso Regression
- Density Estimation
 - MLE
 - MAP

Ridge Regression

- Ridge regression solves the multicollinearity problem through shrinkage parameter λ

$$= \operatorname{argmin}_{\beta \in \mathbb{R}^p} \underbrace{\|y - X\beta\|_2^2}_{\text{Loss}} + \lambda \underbrace{\|\beta\|_2^2}_{\text{Penalty}}$$

- Ridge regression shrinks the value of coefficients but doesn't reaches zero.
- It uses a regularization method, **l2 regularization**.
- **Regularization:** The process of adding a norm to our cost function.

Lasso Regression

- **Lasso:** Least Absolute Shrinkage and Selection Operator.

$$= \operatorname{argmin}_{\beta \in \mathbb{R}^p} \underbrace{\|y - X\beta\|_2^2}_{\text{Loss}} + \lambda \underbrace{\|\beta\|_1}_{\text{Penalty}}$$

- Lasso penalizes the absolute size of the regression coefficients.
- It reduces the variability and improving the accuracy of linear regression models.
- Difference with ridge: Lasso uses absolute values in the penalty function, instead of squares.
- This causes some of the parameter estimates to turn out exactly zero (i.e., variable selection out of given n variables).

ElasticNet Regression

- hybrid of Lasso and Ridge Regression techniques.
- Uses L1 and L2 prior as regularizer.
- Elastic-net is useful when there are multiple features which are correlated. Lasso is likely to pick one of these at random, while elastic-net is likely to pick both.

$$\hat{\beta} = \underset{\beta}{\operatorname{argmin}} (\|y - X\beta\|^2 + \lambda_2 \|\beta\|^2 + \lambda_1 \|\beta\|_1).$$

Bayesian Regression

- Recall simple linear regression:

$$y_i = \alpha + \beta x_i + \varepsilon$$

- Let's write it in terms of the probabilistic model behind it

$$\mu_i = \alpha + \beta x_i$$

$$y_i \sim \mathcal{N}(\mu_i, \sigma)$$

- i.e. Y follows normal distribution parametrized by μ_i (a linear function of X parametrized by α, β), and by standard deviation σ .
- If we estimate using ordinary least squares, no need to consider probabilistic formulation, since we want to find optimal values of α, β by minimizing the squared errors.
- On another hand, we could estimate such model using MLE (maximizing the likelihood function):

$$\arg \max_{\alpha, \beta, \sigma} \prod_{i=1}^n \mathcal{N}(y_i; \alpha + \beta x_i, \sigma)$$

Density Estimation

- density estimation refers to estimating the probability distribution (or density function) of a random variable based on observed data.

Maximum Likelihood Estimation

- **Maximum likelihood estimation (MLE)** is a method of **estimating** the **parameters** of an assumed **probability distribution**, given some observed data.

Bayesian Regression

- MLE:

$$\arg \max_{\alpha, \beta, \sigma} \prod_{i=1}^n \mathcal{N}(y_i; \alpha + \beta x_i, \sigma)$$

- where $\mathcal{N}$ is a density function of normal distribution evaluated at y_i points, parametrized by means $\alpha + \beta x_i$ and standard deviation σ .
- In the Bayesian approach instead of maximizing the likelihood function, we would assume **prior distributions** for the parameters and use the Bayes theorem

$$\text{posterior} \propto \text{likelihood} \times \text{prior}$$

$$\underbrace{f(\alpha, \beta, \sigma | Y, X)}_{\text{posterior}} \propto \underbrace{\prod_{i=1}^n \mathcal{N}(y_i | \alpha + \beta x_i, \sigma)}_{\text{likelihood}} \underbrace{f_{\alpha}(\alpha) f_{\beta}(\beta) f_{\sigma}(\sigma)}_{\text{priors}}$$

Bayesian Regression

posterior $\propto$ likelihood $\times$ prior

$$\arg \max_{\alpha, \beta, \sigma} \prod_{i=1}^n \mathcal{N}(y_i; \alpha + \beta x_i, \sigma)$$

- The likelihood function is as above in MLE, however what changes in Bayesian approach is that we assume some prior distributions for the estimated parameters α, β, σ and include them into the equation

$$\underbrace{f(\alpha, \beta, \sigma | Y, X)}_{\text{posterior}} \propto \underbrace{\prod_{i=1}^n \mathcal{N}(y_i | \alpha + \beta x_i, \sigma)}_{\text{likelihood}} \underbrace{f_{\alpha}(\alpha) f_{\beta}(\beta) f_{\sigma}(\sigma)}_{\text{priors}}$$

- While in MLE we look for a single optimal value for each of the parameters, in the Bayesian approach we obtain the **posterior distribution** of the parameters.

Mathematical Form (Bayes' Theorem)

- Where:
 - $P(\theta|\text{data})$ is the posterior distribution.
 - $P(\text{data}|\theta)$ is the likelihood function.
 - $P(\theta)$ is the prior distribution, which reflects your beliefs about θ before seeing the data.
 - $P(\text{data})$ is a normalizing constant.

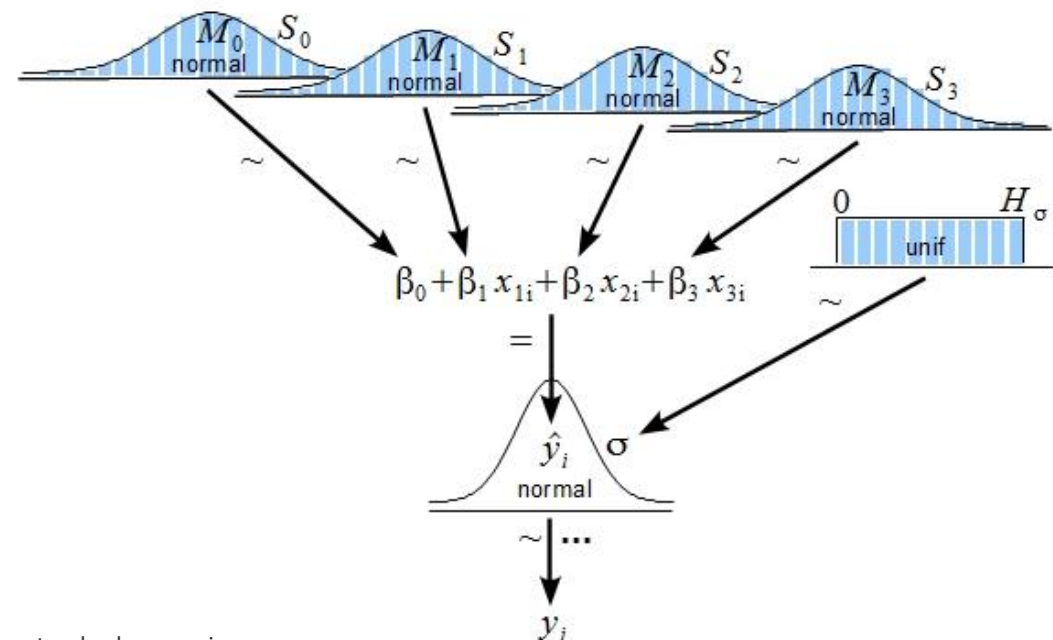
$$P(\theta \mid \text{data}) = \frac{P(\text{data} \mid \theta) \cdot P(\theta)}{P(\text{data})}$$

Bayesian Regression

- What distribution for priors?

- Unlimited number of choices.
- For example, assuming normal distribution for α, β
- For σ assume some prior distribution that is bounded to be greater than zero since σ needs to be positive.

$$\underbrace{f(\alpha, \beta, \sigma | Y, X)}_{\text{posterior}} \propto \underbrace{\prod_{i=1}^n \mathcal{N}(y_i | \alpha + \beta x_i, \sigma)}_{\text{likelihood}} \underbrace{f_\alpha(\alpha) f_\beta(\beta) f_\sigma(\sigma)}_{\text{priors}}$$



Bayesian Regression

- The final estimate of posterior depends on the information that comes from your data and from your priors, but the more information is contained in your data, the less influential are priors.

$$\underbrace{f(\alpha, \beta, \sigma \mid Y, X)}_{\text{posterior}} \propto \underbrace{\prod_{i=1}^n \mathcal{N}(y_i \mid \alpha + \beta x_i, \sigma)}_{\text{likelihood}} \underbrace{f_\alpha(\alpha) f_\beta(\beta) f_\sigma(\sigma)}_{\text{priors}}$$

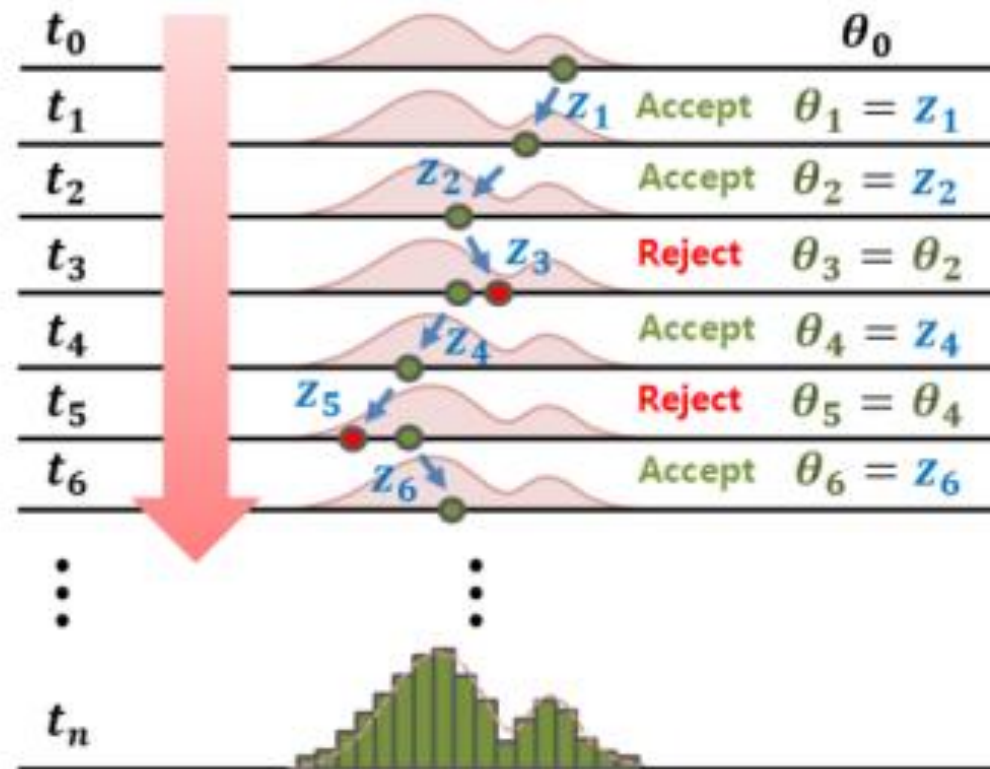
- If using uniform priors, since they are $f(\theta) \propto 1$, so the posterior distribution will reach its maximum at exactly the same point as the maximum likelihood estimate.
- In some cases, we can use **conjugate priors**, so the posterior distribution is directly available.
- However, in most cases, posterior distribution will not be directly available and we need to use Markov Chain Monte Carlo methods for estimating the model.

Conjugate Prior

- What is conjugate prior?
- “If the posterior distribution $p(\theta \mid \text{data})$ is in the same probability distribution family as the prior probability distribution $p(\theta)$, the prior and posterior are then called **conjugate distributions**, and the prior is called a **conjugate prior** for the likelihood function $p(\text{data} \mid \theta)$.”

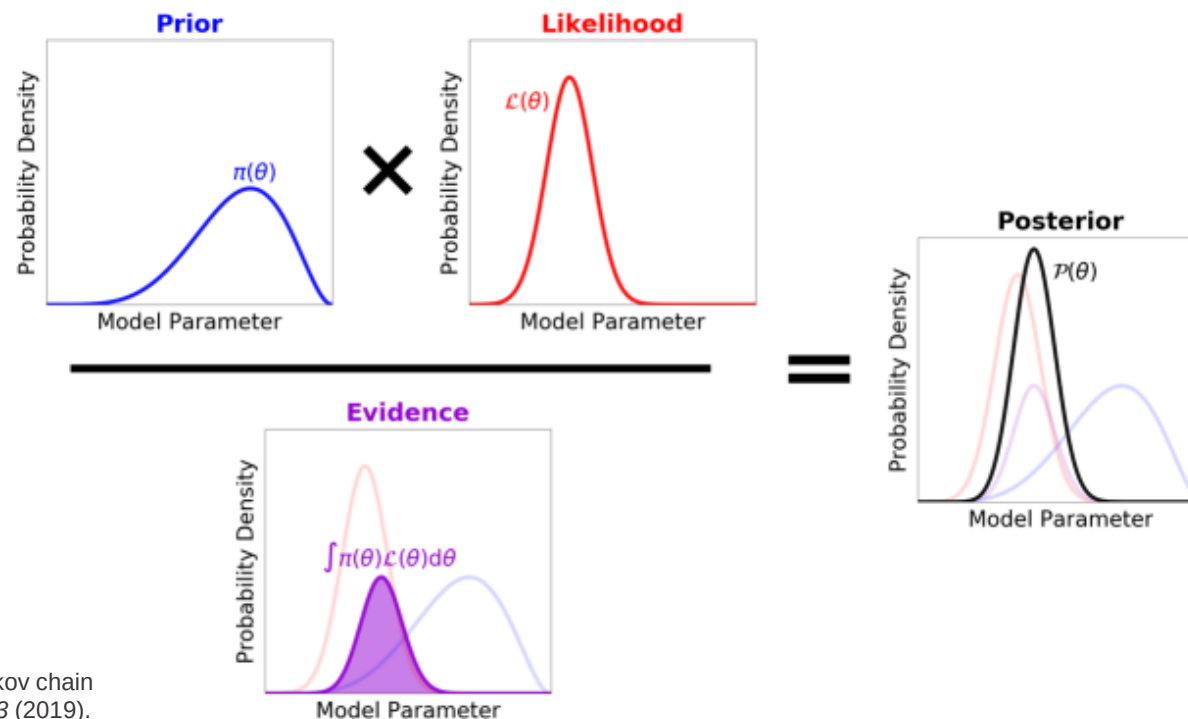
Illustration of Markov Chain Monte Carlo method

- MCMC: a class of algorithms for sampling from a probability distribution.



An illustration of Bayes' Theorem

- The posterior probability $P(\Theta)$ (black) of our model parameters Θ is based on a combination of our prior beliefs $\pi(\Theta)$ (blue) and the likelihood $L(\Theta)$ (red), normalized by the overall evidence $Z = \int \pi(\Theta)L(\Theta)d\Theta$.



Density Estimation

- Maximum likelihood (ML)
- Maximum a posteriori (MAP)
- Bayesian

The importance of probabilistic approach

- When learning some target function $f : X \rightarrow Y$, we can more generally learn the probabilistic function $P(Y|X)$.
- Using probabilistic approach allows us **designing algorithms that learn functions with uncertain outcomes** and incorporating prior knowledge to guide learning.
- Example:
 - Predicting tomorrow's stock price,
 - Prior knowledge: a bias that tomorrow's stock price is likely to be similar to today's price

Joint Probability Distributions

- **The key point in creating probabilistic models** is to define a set of random variables, and to consider the joint probability distribution over them.
- **Example:** a joint probability distribution over three random variables: a person's Gender, the number of HoursWorked each week, and their Wealth.

Gender	HoursWorked	Wealth	probability
female	< 40.5	poor	0.2531
female	< 40.5	rich	0.0246
female	$\geq$ 40.5	poor	0.0422
female	$\geq$ 40.5	rich	0.0116
male	< 40.5	poor	0.3313
male	< 40.5	rich	0.0972
male	$\geq$ 40.5	poor	0.1341
male	$\geq$ 40.5	rich	0.1059

The importance of knowing probability distribution

- Once we know the joint distribution we can answer every possible probabilistic question about these variables.
- if the joint probability distribution over an arbitrary set of random variables $\{X_1 \dots X_n\}$, is known then the conditional and joint probability distributions for arbitrary subsets of these variables (e.g., $P(X_n | X_1 \dots X_{n-1})$) can be calculated.
- This can be used to solve any classification, regression, or other function approximation problem defined over these variables, and furthermore produce probabilistic rather than deterministic predictions.
- **Example:** predict which people are rich or poor based on their gender and hours worked, $P(\text{Wealth} | \text{Gender}, \text{HoursWorked})$.

Learning the Joint Distribution

- How can we learn joint distributions from observed training data?
- Begin with a large database containing, say, descriptions of a million people?
- How the learning problem would change if we were to add additional variables to describe a total of 100 boolean features for each person?
- How many people on earth? approximately 10^{10} people

Learning the Joint Distribution

- To learn probabilities from available training data, we must be smart about
 - (1) how we estimate probability **parameters** from available data, and
 - (2) how we **represent joint probability** distributions.

Estimating Probabilities

- Example: flip a coin
- The coin is represented by the random variable X . If you flip this coin, it may turn up heads (indicated by $X = 1$) or tails ($X = 0$).
- The learning task is to estimate the probability that it will turn up heads; $P(X = 1) = ?$
- θ : the true (but unknown) probability of heads (e.g., $P(X = 1) = \theta$)
- $\hat{\theta}$: our learned estimate of the true θ .
- Let's gather training data by flipping the coin n times, and observe that it turns up heads α_1 times, and tails α_0 times.
- $n = \alpha_1 + \alpha_0$.

Estimating Probabilities

- What is the most intuitive approach to estimating $\theta = P(X = 1)$?

Probability estimation Algorithm 1: Given observed training data producing α_1 total "heads," and α_0 total "tails," output the estimate

$$\hat{\theta} = \frac{\alpha_1}{\alpha_1 + \alpha_0}$$

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This called the maximum likelihood estimate (MLE).

Estimating Probabilities

Example: flip the coin 50 times, observing 24 heads and 26 tails, then what is the estimated probability of $P(X = 1)$? $\hat{\theta} = 0.48$.

- The approach is reasonable and very intuitive.
- A good approach when plenty of training data exist.
- When the training data is scarce it can produce unreliable estimates.

Example: only 3 flips of the coin, observe $\alpha_1 = 1$ and $\alpha_0 = 2$, so $\hat{\theta} = 0.33$.

How would we respond to this?

Estimating Probabilities

- What if we have **prior knowledge** about the coin?
 - for example, a government minted coin (likely to have θ near 0.5)
- Still believing the probability is 0.5 than Algorithm 1 estimate of $\hat{\theta} = 0.33$.
- So, we need a second intuitive algorithm: an algorithm that enables us to incorporate prior assumptions along with observed training data to produce the final estimate.

Estimating Probabilities

- Algorithm 2 allows us to express our prior assumptions or knowledge about the coin by adding in any number of imaginary coin flips resulting in heads or tails.
- γ_1 : imaginary heads,
- γ_0 : imaginary tails.
- **Probability estimation Algorithm 2:** Given observed training data producing α_1 "heads," and α_0 "tails," plus prior information expressed by introducing γ_1 imaginary "heads" and γ_0 imaginary "tails," output the estimate

$$\hat{\theta} = \frac{(\alpha_1 + \gamma_1)}{(\alpha_1 + \gamma_1) + (\alpha_0 + \gamma_0)}$$

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$$\hat{\theta} = \frac{(\alpha_1 + \gamma_1)}{(\alpha_1 + \gamma_1) + (\alpha_0 + \gamma_0)}$$

- This is called maximum a posteriori probability (MAP).

A few points

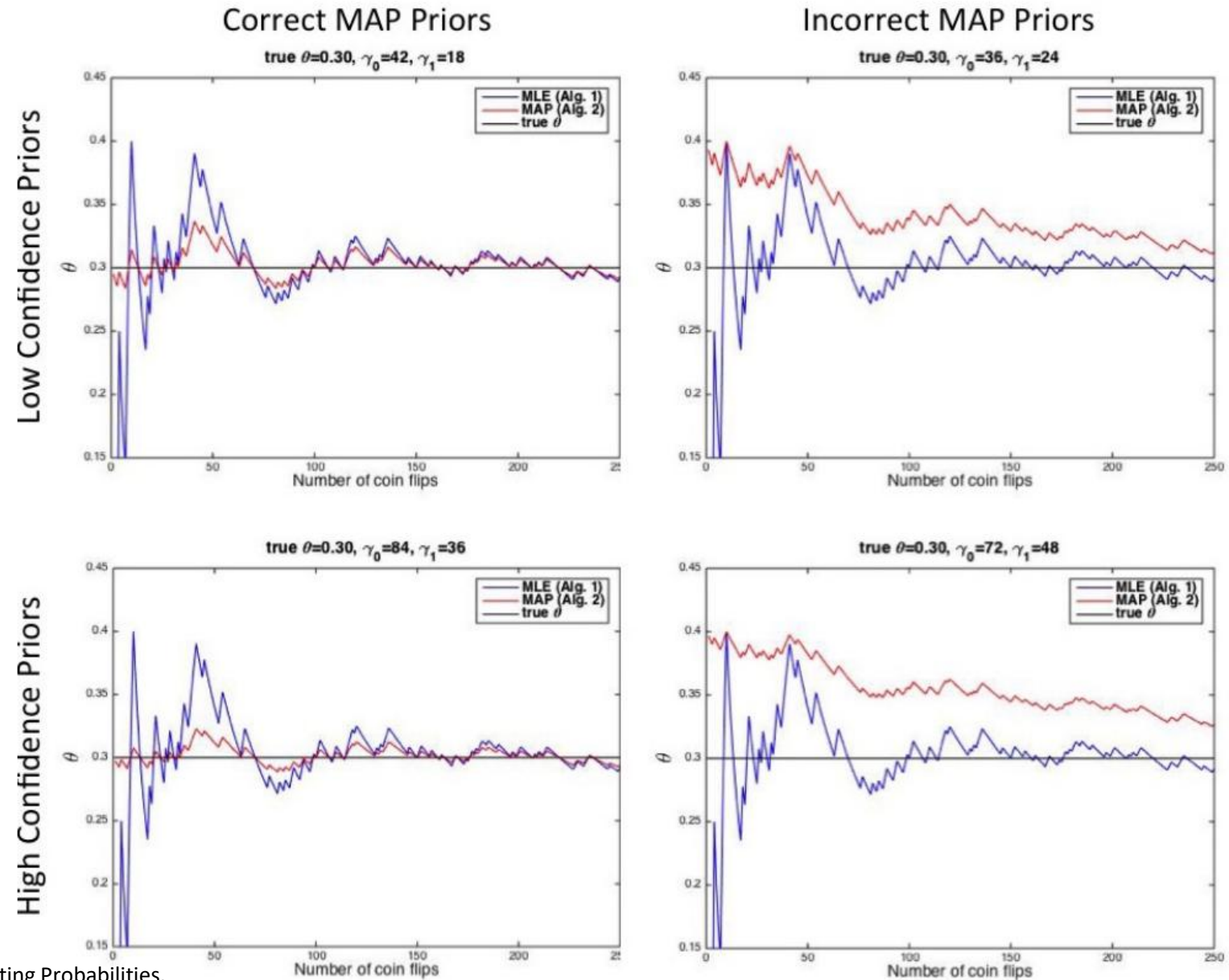
- Incorporate our prior assumptions about the value of θ by adjusting the ratio of γ_1 to γ_0 .
- Express our degree of certainty about our prior knowledge, by adjusting the total volume of imaginary coin flips. (use $\gamma_1 = 700$ and $\gamma_0 = 300$ instead of $\gamma_1 = 7$ and $\gamma_0 = 3$).
- If we set $\gamma_1 = \gamma_0 = 0$, then Alg 2 produces exactly the same estimate as Algorithm 1. Alg 1 is just a special case of Alg 2.
- Asymptotically, as the volume of actual observed data grows toward infinity, the influence of imaginary data goes to 0 (imaginary data becomes insignificant compared to a sufficiently large number of actual observations).
- In Algorithm 2, priors have the strongest influence when observations are scarce.

Algorithm 1 and Algorithm 2

- The two most widely used approaches to machine learning of probabilistic models from training data.
- **Algorithm 1:** Maximum Likelihood Estimation: an estimate of θ that maximizes the probability of the observed data.
- **Algorithm 2:** Maximum a Posteriori (MAP) estimation: the estimate of θ that is itself most probable, given the observed data, plus background assumptions about its value.

The experimental behavior of these two algorithms

MLE and MAP estimates of θ as the number of coin flips grows



Maximum Likelihood Estimation (MLE)

- MLE estimates parameters θ based on the principle that if we observe training data D , we should choose the value of θ that makes D most probable.

$$\hat{\theta}^{MLE} = \arg \max_{\theta} P(D|\theta)$$

- **The underlying principle:** we are more likely to observe data D if we are in a world where the appearance of this data is highly probable. Therefore, we should select θ value that maximizes the probability of having observed D .
- Example: flipping a coin
- X : a random variable which can take on either value 1 or 0, and $\theta = P(X = 1)$
- Flip the coin X a number of times to produce training data D , (we observe $X = 1$ a total of α_1 times, and $X = 0$ a total of α_0 times).

Maximum Likelihood Estimation

- **MLE:** choosing θ to maximize $P(D | \theta)$.
- Write an expression for $P(D | \theta)$, or equivalently $P(\alpha_1, \alpha_0 | \theta)$ in terms of θ , then find an algorithm that chooses θ that maximizes this quantity.
- **Case 1:** if data D consists of just one coin flip, then $P(D | \theta) = \theta$ if $X = 1$, and $P(D | \theta) = (1 - \theta)$ if $X = 0$.
- **Case 2:** if a set of i.i.d. coin flips such as $D = \{1, 1, 0, 1, 0\}$, then we can easily calculate $P(D | \theta)$ by multiplying the probabilities of each individual coin flip

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$$P(D = \{1, 1, 0, 1, 0\} | \theta) = \theta \cdot \theta \cdot (1 - \theta) \cdot \theta \cdot (1 - \theta) = \theta^3 \cdot (1 - \theta)^2$$

Maximum Likelihood Estimation

- Generally, if the total number of observed times α_1 when $X = 1$ and the number of times α_0 that $X = 0$, we have

$$P(D = \langle \alpha_1, \alpha_0 \rangle | \theta) = \theta^{\alpha_1} (1 - \theta)^{\alpha_0}$$

- $P(D | \theta)$ is often called the data likelihood, or the data likelihood function.

$$\arg \max_{\theta} P(D | \theta) = \arg \max_{\theta} \ln P(D | \theta)$$

Maximum Likelihood Estimation

$$\begin{aligned}\frac{\partial \ell(\theta)}{\partial \theta} &= \frac{\partial \ln P(D|\theta)}{\partial \theta} \\ &= \frac{\partial \ln[\theta^{\alpha_1} (1-\theta)^{\alpha_0}]}{\partial \theta} \\ &= \frac{\partial [\alpha_1 \ln \theta + \alpha_0 \ln(1-\theta)]}{\partial \theta} \\ &= \alpha_1 \frac{\partial \ln \theta}{\partial \theta} + \alpha_0 \frac{\partial \ln(1-\theta)}{\partial \theta} \\ &= \alpha_1 \frac{\partial \ln \theta}{\partial \theta} + \alpha_0 \frac{\partial \ln(1-\theta)}{\partial(1-\theta)} \cdot \frac{\partial(1-\theta)}{\partial \theta} \\ \frac{\partial \ell(\theta)}{\partial \theta} &= \alpha_1 \frac{1}{\theta} + \alpha_0 \frac{1}{(1-\theta)} \cdot (-1)\end{aligned}$$

Maximum Likelihood Estimation

$$\begin{aligned}0 &= \alpha_1 \frac{1}{\theta} - \alpha_0 \frac{1}{1-\theta} \\ \alpha_0 \frac{1}{1-\theta} &= \alpha_1 \frac{1}{\theta} \\ \alpha_0 \theta &= \alpha_1 (1-\theta) \\ (\alpha_1 + \alpha_0) \theta &= \alpha_1 \\ \theta &= \frac{\alpha_1}{\alpha_1 + \alpha_0}\end{aligned}$$

References

- Tom M. Mitchell, Machine Learning, 2017, Chapter 2: Estimating Probabilities, http://www.cs.cmu.edu/~tom/mlbook/Joint_MLE_MAP.pdf
- <https://www.javatpoint.com/linear-regression-vs-logistic-regression-in-machine-learning>
- <https://medium.com/mlearning-ai/multiple-linear-regression-fundamentals-and-modeling-in-python-60db7095deff>
- <https://towardsdatascience.com/cross-validation-explained-evaluating-estimator-performance-e51e5430ff85>
- <https://towardsdatascience.com/evaluation-metrics-model-selection-in-linear-regression-73c7573208be>
- IBM Machine Learning with Python: A Practical Introduction